Solving Compound Inequalities

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Learning Objectives

- Distinguish between 'and' (intersection) and 'or' (union) inequalities
- Solve three-part compound inequalities
- Graph compound solution sets on a number line
- Write compound solutions in interval notation

Solve each compound inequality, showing all steps. Decide whether the solution is an intersection ('and') or a union ('or'), then write it in interval notation.

1. Solve and write in interval notation:

$$-3 < 2x + 1 \leq 5$$

Answer: ________________________________

2. Solve the "and" inequality:

$$x - 2 < 3 \text{ and } x + 1 > 0$$

Answer: ________________________________

3. Solve the "or" inequality:

$$2x + 3 < -1 \text{ or } x - 4 \geq 0$$

Answer: ________________________________

4. Solve and graph:

$$-6 \leq 3x < 12$$

Answer: ________________________________

5. Solve. Flip signs when dividing by a negative:

$$1 < 4 - x < 6$$

Answer: ________________________________

6. Solve the "or" inequality:

$$4x + 1 \leq -7 \text{ or } 3x - 2 > 7$$

Answer: ________________________________

7. Solve and write in interval notation:

$$-2 \le \frac{x+1}{2} < 3$$

Answer: ____________________________________

8. Solve the "and" inequality:

$$3x - 1 \ge 5 \quad \text{and} \quad 2x < 14$$

Answer: ____________________________________

9. Solve:

$$0 \le 2x - 4 \le 10$$

Answer: ____________________________________

10. Solve the "or" inequality:

$$x + 5 < 3 \quad \text{or} \quad \frac{x}{2} \ge 1$$

Answer: ____________________________________

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ANSWER KEY & SOLUTIONS

'and' keeps the overlap of the two solution sets; 'or' keeps the union. Flip the sign when dividing all parts by a negative.

Solutions

1. Solve and write in interval notation:

$-3 < 2x + 1 \le 5$

→ *Subtract 1 from all parts: -4 < 2x ≤ 4.*
→ *Divide all by 2: -2 < x ≤ 2.*

Answer: $-2 < x \le 2 \quad \Rightarrow \quad (-2,\ 2]$

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2. Solve the "and" inequality:

$x - 2 < 3 \;\text{ and }\; x + 1 > 0$

→ *First: x < 5.*
→ *Second: x > -1.*
→ *"and" = overlap: -1 < x < 5.*

Answer: $-1 < x < 5 \quad \Rightarrow \quad (-1,\ 5)$

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3. Solve the "or" inequality:

$2x + 3 < -1 \;\text{ or }\; x - 4 \ge 0$

→ *First: 2x < -4, so x < -2.*
→ *Second: x ≥ 4.*
→ *"or" = union of both pieces.*

Answer: $x < -2 \;\text{ or }\; x \ge 4$

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4. Solve and graph:

$-6 \le 3x < 12$

→ *Divide all three parts by 3: -2 ≤ x < 4.*
→ *Closed at -2, open at 4.*

Answer: $-2 \le x < 4 \quad \Rightarrow \quad [-2,\ 4)$

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5. Solve. Flip signs when dividing by a negative:

$1 < 4 - x < 6$

→ *Subtract 4: -3 < -x < 2.*
→ *Multiply all by -1 and FLIP: 3 > x > -2.*
→ *Rewrite: -2 < x < 3.*

Answer: $-2 < x < 3 \quad \Rightarrow \quad (-2,\ 3)$

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6. Solve the "or" inequality:

$$4x + 1 \leq -7 \text{ or } 3x - 2 > 7$$

→ *First: 4x ≤ -8, so x ≤ -2.*
→ *Second: 3x > 9, so x > 3.*
→ *Union of both.*

Answer: $x \leq -2 \text{ or } x > 3$

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7. Solve and write in interval notation:

$$-2 \leq \frac{x+1}{2} < 3$$

→ *Multiply all by 2: -4 ≤ x + 1 < 6.*
→ *Subtract 1: -5 ≤ x < 5.*

Answer: $-5 \leq x < 5 \quad \Rightarrow \quad [-5, 5)$

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8. Solve the "and" inequality:

$$3x - 1 \geq 5 \text{ and } 2x < 14$$

→ *First: 3x ≥ 6, so x ≥ 2.*
→ *Second: x < 7.*
→ *Overlap: 2 ≤ x < 7.*

Answer: $2 \leq x < 7 \quad \Rightarrow \quad [2, 7)$

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9. Solve:

$$0 \leq 2x - 4 \leq 10$$

→ *Add 4 to all parts: 4 ≤ 2x ≤ 14.*
→ *Divide by 2: 2 ≤ x ≤ 7.*

Answer: $2 \leq x \leq 7 \quad \Rightarrow \quad [2, 7]$

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10. Solve the "or" inequality:

$$x + 5 < 3 \text{ or } \frac{x}{2} \geq 1$$

→ *First: x < -2.*
→ *Second: x ≥ 2.*
→ *Union of both pieces.*

Answer: $x < -2 \text{ or } x \geq 2$